ESP32 + Virtual NTC Sensor

Prepared example to explore firmware behavior. No Wokwi execution evidence is claimed.

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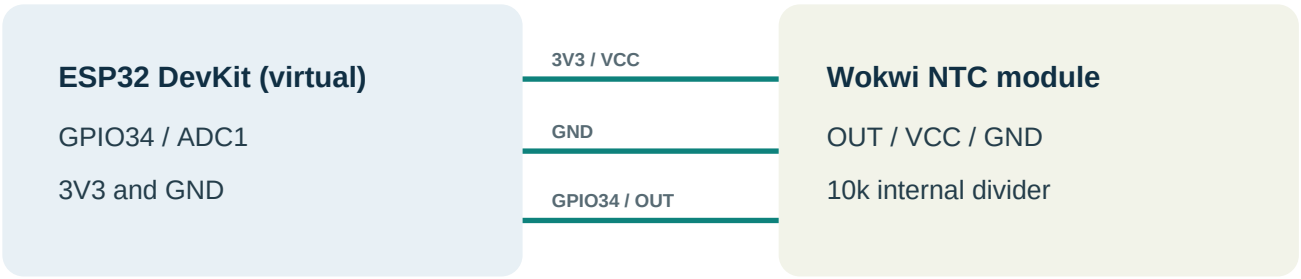
PURPOSE

Exercise GPIO34 ADC acquisition and thermistor conversion with a movable virtual temperature control. Serial output has ms,mv,temp_c,status columns. Useful for exploring firmware thresholds before a physical test.

VIRTUAL CONNECTIONS



WHY THIS IS NOT AN EXACT SCHEMATIC SIMULATION

Wokwi uses an NTC module with a built-in 10k resistor. The real KiCad front end has R1 = 9.82k, R2 = 990 ohms and C1 = 100 nF. Wokwi does not accurately model this complete analog network; LTspice supplies the circuit and filter analysis.

SIMULATION STATUS

diagram.json and sketch.ino are included as an editable starter project.
The simulator has not been run or verified from this environment.
Do not present its expected outcomes as observed Wokwi test results.

Virtual test procedure and handoff

What to do next when you open the included Wokwi project.

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RUN IN A BROWSER

1. Open https://wokwi.com/projects/new/esp32
2. Paste the provided sketch.ino and diagram.json into that project.
3. Run simulation, then open the serial monitor.
4. Change NTC temperature: 25, 35, 45, 50, and 60 degrees C.
5. Record time, calculated temperature, status and any discrepancies.
6. Tag any exported samples SYNTHETIC; do not treat them as hardware.

CHECKLIST (TO BE COMPLETED AFTER AN ACTUAL WOKWI RUN)

- [ ] Sketch compiles in the Wokwi ESP32 environment
- [ ] GPIO34 reads simulated NTC output
- [ ] Serial output uses four defined CSV columns
- [ ] Nominal report interval is approximately 200 ms
- [ ] Status transitions NORMAL / HOT / OUT_OF_RANGE are observed

INTERVIEW EXPLANATION (AFTER TESTING)

"I used Wokwi to exercise the ESP32 acquisition logic with a virtual NTC.
I used LTspice for the actual analog filter and divider model, and
real breadboard measurements for the hardware validation."

EVIDENCE TO PRESERVE

Run screenshot, exported serial log, settings, date and observed status changes.
These were not in the current uploaded files and have not been fabricated.